Tactix proof repair and readiness checkpoint

Codex for Andrew

25 September 2026

Outcome. The AM442 Claude Code campaign, run explicitly with Opus, finished its online-configuration affected scope successfully across 750,000 candidate executions. It also exposed a pre-existing native-LRAT proof defect in ordinary solving. Tactix Sweep had reused a derivation prefix whose hints had been normalized for a different unit target. Commit `421a38a1` retains the raw prefix and normalizes it separately for each target. The old exact proof rejects under CakeLPR; the patched reduced 28-clause and original 129-clause H1 proofs both return `s VERIFIED UNSAT`. The focused Sweep test and the complete Sweep DRAT/LRAT/native-DSR proof suite pass. This is a scoped repair, not combined N2 acceptance. The AM444 Opus owner is preparing its affected fuzz; full Debug job 1723644 is queued on the committed source.

Canonical H2 proof. The public X7 action remains legal. The canonical route uses no-option `dsr-trim`, then LSR and Trestle. On the reduced 16-clause H2 input, moving one deletion before the first derived unit produces a proof verified by both checker directions and Trestle. On the original 42-clause replay, moving all six retained-reason deletions ahead of the first unit yields a backward elaboration independently verified by Trestle, but forward checking rejects the first unit. An exhaustive 64-subset control then found that moving only three selected deletions lets both checker directions and Trestle verify the original proof. It exposes the proof dependencies an emitter must respect; choosing those lines for one CNF is not a general repair. AM443's two-variable negative control shows a reason can already be implied by the original input, leaving no earlier proof-line window. The diagnostic implied unit deletion option changes which proofs verify in both directions, so it has not been adopted as a checker contract.

Reference rule coverage. The AM446 source crosswalk maps the nonincremental CaDiCaL proof producers and SC24 Kissat/AE-Kissat-MAB clausal producer families to Tactix's typed actions. Their emitter interfaces expose RUP, RAT, deletion and finite substitution witnesses; this static mapping does not prove public reachability or correct current certificates. Satsuma's VeriPB mode and selected LymphoSAT winner paths use general pseudo-Boolean arithmetic, redundancy/dominance and subproofs. Tactix has terminal startup PBP for Lucky 19/20 but no general live PB action or proof interface. Particular winner formulas have independently verified Tactix clausal/startup proofs; the general interface gap remains a deliberate architecture question.

Gates and learning readiness. The paired AES diagnostic is terminal: both original-CNF native DSR proofs verify after 5,780/5,790 checker seconds. Its frozen matched Tier-C pair is still unscorable at the 4,800-second checker limit. The older full Debug job 1723391 completed 186/187: its only failure was a stale static classification of Sweep's lifecycle state; the final scripted demo passed. The current tree passes that audit, and fresh Debug job 1723644 is pending. AM427

Sweep lifecycle Opus arrays are live. No current combined Rank 1–50, B28+C5, WV100, WV2-89, N4 reference-parity or Tier-D result follows from these scoped checks. The existing online configuration supports prototyping dose and order, but production learning still needs a shared adjustable inprocessing controller and owned live switches for branching and clause deletion. The final readiness verdict must follow the correctness, fuzz and performance gates.

Next actions. Collect the two Debug suites and both Opus owners, repair any bounded handoff, and solve H2 without suppressing legal public X7 actions. Then check one combined pin against independent proofs/models, the required random-policy and default-policy fuzzing, and the full performance and reference-parity gates. Only after N1–N4 and the final learning-readiness review pass should the 1,166-instance Tier D comparison launch at 5,000 seconds, four CPUs and 32 GiB per instance.

Evidence. `benchmarks/reports/2026-09-25-am444-sweep-lrat-core-target-repair.md`; `benchmarks/reports/2026-09-25-am444-h2-early-deletion-control.md`; `benchmarks/reports/2026-09-25-am445-h2-original-deletion-order-control.md`; `benchmarks/reports/2026-09-25-am448-h2-deletion-dependency-control.md`; `benchmarks/reports/2026-09-25-am447-full-debug-terminal-triage.md`; `benchmarks/reports/2026-09-25-am446-reference-proof-producer-crosswalk.md`; the AM442 and AM427 owner status/report files under `benchmarks/runs/`.